EAMHP

Excel-Based Avionics Mechanical & Health Prediction

LOGIC MANUAL

Rule-Based Fault Detection Engine — V1.1.0

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Author	Nikhilesh (Nikhilesh Avionics Projects)
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1. Overview

The EAMHP Logic Manual documents the internal rule-based decision engine that drives fault detection and maintenance prediction within the EAMHP system. It describes how raw parameter values are transformed into actionable maintenance outputs through a structured, deterministic logic pipeline.

> 📌 **Purpose of This Manual**
>
> This document is intended for AME students, avionics trainees, and technical reviewers who need to understand how EAMHP arrives at a fault prediction — not just what the output is, but exactly why the logic produces it.

EAMHP does not use machine learning or adaptive algorithms. Every fault prediction is the result of explicit, human-readable rules built from Airbus A320 AMM/TSM knowledge and standard avionics maintenance practice. This means:

- Every output is fully explainable and traceable to a specific rule condition
- The system produces the same result for the same input, every time
- Rules can be inspected, validated, and updated by any qualified AME
- No training data, internet connection, or external model is required

2. Logic Philosophy

The EAMHP logic engine is grounded in the same diagnostic methodology used by AMEs on the line — observe a parameter, compare it to known limits, identify the deviation pattern, and consult the appropriate maintenance procedure.

Design Principle	Implementation in EAMHP	Benefit
Parameter Threshold Basis	All evaluations compare live values to certified AMM tolerance bands stored in ReferenceData	Directly aligned with aircraft maintenance standards
Deterministic Logic	Rule-based IF/THEN conditions with no randomness or adaptive behaviour	100% reproducible, auditable outputs
Multi-Parameter Correlation	Fault detection considers combinations of 2 and 3 simultaneous parameter deviations	Captures complex real-world fault signatures
Fault Combination Mapping	680 pre-defined fault ID combinations across 5 fault categories	Comprehensive coverage of A320 engine fault space
Dual-Engine Coverage	Separate logic branches for CFM56-5B and IAE V2500-A5 variants	Accurate engine-specific AMM/TSM procedure output

Design Principle	Implementation in EAMHP	Benefit
Human-Readable Rules	All logic built in Excel formulas and VBA — inspectable by any AME	Transparent, maintainable, and auditable system

3. Logic Structure

The EAMHP logic pipeline operates in five sequential stages. Each stage transforms the data into a more refined form, ultimately producing a structured maintenance recommendation.

3.1 Stage 1 — Parameter Evaluation

Each engine performance parameter entered by the user is evaluated against its certified tolerance band. The tolerance band is retrieved automatically from the ReferenceData sheet using the entered Parameter Name and Engine Type as lookup keys.

> ⚠️ **Tolerance Band Source**
> All tolerance values in EAMHP are derived from Airbus A320 Family AMM under ISA Standard Atmosphere / Sea Level conditions. They represent certified operating limits for the CFM56-5B and IAE V2500-A5 engine variants.

The evaluation logic produces one of three condition states per parameter:

Condition State	Evaluation Rule	Meaning	Fault Status
LOW	Entered Value < Lower Tolerance	Parameter reading is below the minimum certified limit	FAULT (LOW)
NORMAL	Lower Tolerance ≤ Value ≤ Upper Tolerance	Parameter reading is within the certified operating band	NORMAL
HIGH	Entered Value > Upper Tolerance	Parameter reading exceeds the maximum certified limit	FAULT (HIGH)
INVALID	No reference match found for Parameter + Engine Type	Entry does not correspond to a known parameter record	#N/A

3.2 Stage 2 — Condition Mapping

Once each parameter is evaluated, its result is mapped to a logical state label. This abstraction layer converts the numeric comparison into a symbolic state that can be processed by the combination logic in Stage 3.

Condition mapping per parameter group:

Parameter	LOW	NORMAL	HIGH	Indicative Fault Signature
N1 Speed	Compressor underspeed / stall risk	Normal idle / max speed	Fan / LPC overspeed	Performance anomaly / FADEC issue
N2 Speed	HPC underspeed / slow acceleration	Normal HPC operation	HPC overspeed	Turbine / FADEC control fault
EGT	Cold section / fuel starvation sign	Normal combustion temp	Hot section exceedance	Overtemperature — borescope required
Oil Pressure	Oil pump failure / severe leak	Normal lubrication	Blockage / relief valve fault	Oil system malfunction
Oil Temperature	Cold oil / insufficient warm-up	Normal thermal state	Oil cooler failure / overload	Oil system malfunction
Oil Quantity	Low oil level / active leak	Normal reservoir level	Overfill / sensor error	Oil system / servicing issue
Fuel Flow	Fuel starvation / metering unit fault	Normal fuel delivery	Fuel control overfuelling	Compressor / combustion fault
Max Thrust	Thrust deficit / performance loss	Rated thrust output	Thrust exceedance risk	Performance anomaly
Starter Limit	Starter duty cycle within limits	Normal start sequence	Starter duty exceeded	Start system / motoring fault

3.3 Stage 3 — Combination Logic

The EAMHP combination logic is the most critical stage in the pipeline. It correlates multiple simultaneous parameter states to identify compound fault signatures that would not be detectable from a single parameter deviation alone.

> 🧠 **Why Combination Logic Matters**
>
> In real avionics maintenance, most faults manifest as patterns across multiple parameters simultaneously. For example, an oil pump failure shows Low Oil Pressure AND rising Oil Temperature together. EAMHP is designed to detect exactly these compound signatures, mirroring how a qualified AME troubleshoots on the line.

3.3.1 Two-Parameter Combination Logic (F0001–F0120)

When exactly two parameters are simultaneously in a FAULT state, EAMHP cross-references the [Parameter A + Parameter B] pair against the FaultMatrixEngine to retrieve a Fault_ID. There are 120 unique 2-parameter fault combinations in V1.1.0.

Live rule examples from the EAMHP fault matrix:

```
IF  N1_Idle = FAULT  AND  EGT_Idle = FAULT
THEN  Fault_ID: F0004  |  Category: Engine Overtemperature
SEVERITY  HIGH — Immediate borescope inspection required
```

```
IF  N1_Idle = FAULT  AND  Oil_Pressure_Min = FAULT
THEN  Fault_ID: F0007  |  Category: Oil System Malfunction
SEVERITY  HIGH — Inspect oil pump, filter, and cooler
```

```
IF  N1_Idle = FAULT  AND  Fuel_Flow_Cruise = FAULT
THEN  Fault_ID: F0013  |  Category: Compressor Efficiency Loss
SEVERITY  MEDIUM — Perform compressor wash, inspect IGVs and blades
```

```
IF  N1_Idle = FAULT  AND  N1_Max = FAULT
THEN  Fault_ID: F0001  |  Category: Engine Performance Anomaly
SEVERITY  MEDIUM — Engine performance check, FADEC/sensor inspection
```

3.3.2 Three-Parameter Combination Logic (F0121–F0680)

When three parameters are simultaneously in a FAULT state, EAMHP processes the [Parameter A + Parameter B + Parameter C] combination. There are 560 unique 3-parameter fault combinations in V1.1.0, covering more complex fault signatures.

Live rule examples from the EAMHP 3-parameter fault matrix:

```
IF  N1_Idle = FAULT  AND  N1_Max = FAULT  AND  Fuel_Flow_Cruise =
FAULT
THEN  Fault_ID: F0132  |  Category: Compressor Efficiency Loss
SEVERITY  HIGH — LPC/HPC blade inspection + compressor wash + IGV check
```

```
IF  N1_Idle = FAULT  AND  N2_Idle = FAULT  AND  Fuel_Flow_Cruise
= FAULT
THEN  Fault_ID: F0145  |  Category: Compressor Efficiency Loss
SEVERITY  HIGH — Full compressor section inspection, FOD check, wash
```

```
IF  N1_Idle = FAULT  AND  EGT_Idle = FAULT  AND  Fuel_Flow_Cruise
= FAULT
THEN  Fault_ID: F0157  |  Category: Combustion Inefficiency
SEVERITY  CRITICAL — Combustor borescope, fuel nozzle and dome inspection
```

3.4 Stage 4 — Fault Mapping

Once a Fault_ID is identified by the combination logic, EAMHP maps it to a complete fault record containing all information required for the AME to take action. This mapping is performed against the EAMHP fault matrix reference dataset.

Output Field	Data Type	Source Sheet	Description
Fault_ID	String (e.g., F0007)	FaultMatrixEngine	Unique identifier for the matched parameter combination
Parameters_Involved	String list	FaultMatrixEngine	Comma-separated list of all faulted parameters triggering this fault
Fault_Category	Enum (5 categories)	Mixed Report / Category Sheets	Classified fault type for maintenance decision routing
Corrective_Action	Text	Mixed Report	Immediate maintenance action to be taken by the AME
Preventive_Action	Text	Mixed Report	Scheduled maintenance to prevent fault recurrence
CFM56_5B_Procedure	Text	Mixed Report / Engine Sheets	AMM/TSM procedure specific to CFM56-5B engine variant
V2500_Procedure	Text	Mixed Report / Engine Sheets	AMM/TSM procedure specific to IAE V2500-A5 engine variant

3.4.1 Fault Category Routing Logic

The Fault_Category field drives which maintenance decision pathway is presented to the AME. Each category has a defined response protocol:

Fault Category	Trigger Parameters	Immediate Response	AMM Reference
Engine Performance Anomaly	N1, N2, Max Thrust, Starter Limit	Engine performance check, FADEC/sensor inspection	AMM 72-00 / TSM
Engine Overtemperature	EGT Idle, EGT Max TOGA, EGT Max Continuous	Reduce thrust, hot section borescope, fuel schedule check	AMM 72-00 / TSM
Oil System Malfunction	Oil Pressure, Oil Temp, Oil Quantity	Oil pump, filter, cooler inspection; leak check	AMM 79-00 / TSM
Compressor Efficiency Loss	N1/N2 + Fuel Flow Cruise	Compressor wash, IGV/VSV check, FOD inspection	AMM 72-30 / TSM
Combustion Inefficiency	EGT + Fuel Flow + N1/N2	Combustor borescope, fuel nozzle and dome inspection	AMM 72-40 / TSM

3.5 Stage 5 — Output Generation

The final stage assembles the complete maintenance recommendation from the fault mapping data and presents it to the AME through the appropriate category sheet. Output is generated per faulted parameter set and covers all fields needed to initiate a maintenance action.

✅ Output Completeness Guarantee

EAMHP guarantees that 100% of the 680 defined Fault_IDs have non-null outputs for all five fields: Corrective_Action, Preventive_Action, Fault_Category, CFM56-5B Procedure, and V2500 Procedure. No fault ID reaches the output stage without a complete maintenance record.

Complete output record example — Fault F0007 (N1 Idle + Oil Pressure Min):

Output Field	Value
Fault_ID	F0007
Parameters_Involved	N1 Idle, Oil Pressure Min
Fault_Category	Oil System Malfunction
Corrective_Action	Inspect oil pump, check oil filter and cooler, inspect for leaks, verify oil quantity and pressure.
Preventive_Action	Regular oil analysis, scheduled oil filter replacement, periodic oil system inspection.
CFM56_5B_Procedure	CFM56-5B: Inspect oil pump, oil filter, and oil cooler; check scavenge system and leaks.
V2500_Procedure	V2500: Inspect lubrication pump, oil filter bypass valve, verify oil cooler performance and check for leaks.

4. Logic Types Used in EAMHP

EAMHP implements three tiers of logic complexity, each building on the previous to handle increasingly sophisticated fault patterns:

Logic Type	Fault ID Range	Count	Parameter Count	Use Case
Single-Parameter Logic	N/A (baseline evaluation)	—	1	Identifies isolated parameter deviations; populates Fault Status in InputData
Dual-Parameter Combination	F0001 – F0120	120	2	Detects paired fault signatures — primary diagnostic logic layer
Multi-Parameter Combination	F0121 – F0680	560	3	Detects compound fault signatures involving three concurrent deviations

5. Limitations

EAMHP V1.1.0 is a powerful diagnostic support tool, but it operates within clearly defined boundaries. Understanding these limitations is essential for any AME or trainee using the system.

> ⚠️ **Important Notice**
> EAMHP is a decision-support tool for training and maintenance planning assistance. It does not replace official Airbus AMM/TSM documentation, Type Rated AME judgement, or any regulatory airworthiness approval process. Always verify outputs against current AMM revision before taking maintenance action.

Limitation	Impact	Planned Resolution
Static rule-based logic only	Cannot adapt to previously unseen fault patterns not in the rule set	V2.0.0 roadmap: AI/ML-based anomaly detection layer
No adaptive learning	System does not improve from historical fault data or user corrections	V2.0.0 roadmap: Historical data feedback loop
Predefined threshold dependency	Accuracy depends entirely on AMM tolerance values; non-ISA conditions may reduce reliability	User guidance to apply ISA corrections where required

Limitation	Impact	Planned Resolution
Maximum 3 simultaneous parameters	Faults involving 4+ concurrent parameter deviations are not covered in V1.1.0	V1.0.1 planned: 4-parameter combination expansion
No real-time data feed	All parameter values must be manually entered by the user; no ACARS/QAR integration	V1.2.0 roadmap: ACARS data import module
16 parameters only (V1.1.0)	Vibration, fuel flow idle, hydraulic, and avionics parameters not yet included	V1.0.1: Expanded parameter set planned
Windows Excel only	Not compatible with Google Sheets or macOS Excel due to VBA dependency	Long-term: Web app migration under evaluation

6. Logic Coverage Summary

Fault Category	Total Fault IDs	CFM56-5B	IAE V2500-A5	Logic Stage
Engine Performance Anomaly	**45**	✔	✔	Stages 3–4
Engine Overtemperature	**85**	✔	✔	Stages 3–4
Oil System Malfunction	**497**	✔	✔	Stages 3–4
Compressor Efficiency Loss	**11**	✔	✔	Stages 3–4
Combustion Inefficiency	**42**	✔	✔	Stages 3–4

TOTAL: 680 Fault Combinations — 5 Categories — 2 Engine Variants — 16 Parameters — 100% Action Coverage